

Related Rates

ANSWER KEY



Setting up related-rates problems

- 1 A circular oil spill is expanding so that its radius increases at 3 m/min. How fast is the area increasing when the radius is 50 m?

$$A = \pi r^2, \text{ so } \frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(50)(3) = 300\pi \approx 942.5 \text{ m}^2/\text{min}.$$

- 2 A 10 m ladder leans against a wall. The bottom slides away from the wall at 2 m/s. How fast is the top sliding down when the bottom is 6 m from the wall?

$$x^2 + y^2 = 100. \text{ Differentiating: } 2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0. \text{ At } x = 6, y = 8: 12(2) + 16 \frac{dy}{dt} = 0, \text{ so } \frac{dy}{dt} = -1.5 \text{ m/s} \\ \text{(the top is sliding down at 1.5 m/s).}$$

Volume and other rates

- 3 Air is pumped into a spherical balloon at 100 cm³/s. Find the rate of change of the radius when $r = 5$ cm, to three decimal places.

$$V = \frac{4}{3}\pi r^3, \text{ so } \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}. 100 = 4\pi(25) \frac{dr}{dt}, \text{ so } \frac{dr}{dt} = \frac{1}{\pi} \approx 0.318 \text{ cm/s}.$$

- 4 Water drains from a conical tank (vertex down) with radius 4 m and height 10 m at the top, at a rate of 2 m³/min. Using similar triangles $r = \frac{2h}{5}$, find how fast the water level is dropping when $h = 5$ m.

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \left(\frac{2h}{5}\right)^2 h = \frac{4\pi}{75} h^3. \text{ So } \frac{dV}{dt} = \frac{12\pi}{75} h^2 \frac{dh}{dt}. \text{ At } h = 5: -2 = \frac{12\pi}{75}(25) \frac{dh}{dt}, \text{ giving} \\ \frac{dh}{dt} = -\frac{2 \times 75}{300\pi} = -\frac{1}{2\pi} \approx -0.159 \text{ m/min}.$$